ROHIT JHA D

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with strong foundation in end-to-end Machine Learning, Deep Neural Architectures, and Generative AI frameworks. Experienced in deploying enterprise predictive analytics and automated data pipelines at HPCL and formulating unsupervised deep learning models (PyTorch, Deep Autoencoders) at C-DAC. Proven proficiency in NLP, Computer Vision, vector embeddings (FAISS), LangChain orchestration concepts, and containerized REST API deployment (FastAPI/Docker) optimized for scalable production serving.

TECHNICAL SKILLS

- **Core AI / ML & Deep Learning:** Python, PyTorch, TensorFlow, Keras, Scikit-learn, Regression, Classification, Clustering, XGBoost, Deep Autoencoders, Isolation Forest
- **Generative AI & Multimodal:** Generative AI, Large Language Models (LLMs), Prompt Engineering, LangChain, Vector Databases (FAISS, pgvector), NLP, Computer Vision, CLIP
- **Data Engineering & Analysis:** NumPy, Pandas, SQL (PostgreSQL, MySQL), Data Cleansing, Feature Engineering, Exploratory Data Analysis (EDA), Matplotlib, Seaborn
- **Deployment & DevOps:** REST APIs, FastAPI, Flask, Docker, Git/GitHub, Linux/Bash, Model Evaluation & Optimization (AUC-ROC, F1, RMSE, Latency)

PROFESSIONAL EXPERIENCE

Hindustan Petroleum Corporation Limited (HPCL) — **Trichy, India | Apr 2024 – Mar 2025**

Data Science & Analytics Trainee – Predictive Operations (NATS Apprenticeship)

- Developed automated ETL pipelines using Python and SQL across SAP ERP, CRIS, and TIBCO Spotfire, standardizing multi-source supply chain datasets for machine learning modeling.
- Formulated statistical risk scoring models across INR 90+ Crore in B2B accounts, predicting default likelihood on overdue receivables (DV >15 days) to safeguard working capital.
- Applied time-series clustering and variance modeling across 1,400+ fuel retail stations to forecast dryout likelihood and optimize supply replenishment schedules.
- Automated analytical computation pipelines and KPI tracking dashboards in Python and SQL, cutting recurring reporting turnaround time by 35% across 5 sales territories.

Centre for Development of Advanced Computing (C-DAC) — **Chennai, India | Jan 2023 – Jun 2023**

Machine Learning Research Intern – Deep Learning & Anomaly Modeling

- Formulated and implemented unsupervised deep learning architectures (Deep Autoencoders and Isolation Forest) in PyTorch to detect multivariate anomalies across 500,000+ daily log sequences.
- Engineered robust feature transformation pipelines using Python, scaling and extracting latent behavioral embeddings to reduce false discovery rate by 40%.
- Optimized model evaluation metrics, improving anomaly detection recall by 25% on holdout validation data via reconstruction error threshold calibration.
- Benchmarked latent representation learning against classical statistical baselines, documenting model drift characteristics and inference latency for production deployment.

KEY AI / MACHINE LEARNING PROJECTS

Multimodal AI Semantic Search Engine | PyTorch, OpenAI CLIP, Streamlit, Vector Retrieval

- Constructed a zero-shot cross-modal retrieval system projecting natural language prompts and image assets into a joint 512-dimensional embedding space using OpenAI CLIP.
- Implemented dense vector indexing via FAISS and Cosine Similarity computations, achieving sub-second Top-K nearest-neighbor semantic search retrieval across unstructured datasets.

Credit Card Transaction Fraud Detection | Python, Pandas, Scikit-learn, Flask, Docker

- Trained and cross-validated a Random Forest classifier addressing severe transaction class imbalance (14.6% fraud ratio), achieving 98.4% accuracy, 0.97 precision, and 0.96 F1-score.
- Packaged the trained model into a containerized REST API with strict input payload validation, delivering sub-50ms inference latency for automated real-time transaction scoring.

Vehicle Telemetry Regression & Analytics Service | Scikit-learn, Python, Flask, REST API

- Conducted end-to-end exploratory data analysis, correlation profiling, and feature scaling across multi-sensor engine telemetry records.
- Trained and cross-validated an ensemble Random Forest Regressor reaching an R^2 score of 0.91 with minimized root-mean-square error (RMSE).

EDUCATION

Bachelor of Technology (B.Tech.) in Information Technology — **Aug 2019 – Jul 2023**

Anna University (Gojan School of Business and Technology), Chennai | CGPA: 8.42 / 10.0

CERTIFICATIONS

- **Google Data Analytics Professional Certificate –** Coursera / Google (SQL, R, Tableau, Spreadsheets, Data Cleaning, Statistical Foundations)
- **Graduate Apprenticeship Training Completion Certificate (NATS) –** Ministry of Education, Govt. of India (HPCL Trichy)